



Improving Cross-Attention based on Positional Alignment during Inference for Robust Long-form Speech Recognition

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Abstract

End-to-end (E2E) models have significantly advanced automatic speech recognition (ASR), with hybrid architectures that combine Connectionist Temporal Classification (CTC) and attention-based encoder-decoder (AED) models demonstrating superior performance. However, AED architectures, particularly Conformer, face notable challenges with long-form speech, with performance degradation becoming evident for audio exceeding 25 seconds. In this study, we propose improving the Conformer's robustness for long-form ASR by applying Gaussian masking to the cross-attention mechanism of the Transformer decoder during inference, using the aligned positions obtained from the CTC prefix score. The proposed method achieves an error reduction rate (ERR) of 88.41% (from 26.41% to 3.06%) for audio longer than 20 seconds on a LibriSpeech evaluation set constructed by concatenating three utterances. Moreover, the method remains effective with either a Transformer or an E-Branchformer encoder.

Index Terms: Cross-attention, hybrid CTC/attention, Conformer, Long-form speech, ASR

1. Introduction

With the advent of end-to-end (E2E) models, automatic speech recognition (ASR) has seen remarkable advancements. Connectionist Temporal Classification (CTC) [1] and Attention-based encoder-decoder (AED) models [2, 3] have played pivotal roles in the evolution of E2E ASR systems. Subsequently, the hybrid CTC/attention architecture [4], combining CTC and AED models, has been proposed, demonstrating superior ASR performance compared to previous models.

Recent work has focused on improving encoder architectures, which can be seamlessly applied into hybrid structures. Transformer, which achieved significant success in machine translation [5], has also proven effective in ASR [6], demonstrating an exceptional ability to capture global context through self-attention mechanisms. Conformer surpasses the Transformer by incorporating both self-attention and convolutional modules [7], enabling the learning of both local and global contexts, making it one of the most prominent encoder models. E-Branchformer [8, 9] achieves state-of-the-art performance by enhancing the merge module of the Branchformer [10], which captures local and global context through two separate branches.

However, conventional AED models exhibit significant performance degradation in long-form speech [11, 12, 13]. Especially, the Conformer with hybrid architecture has shown weaknesses in long-form exceeding 25 seconds [14]. Conventional ASR models have primarily focused on short-form speech [11], leading to the development of methods that process long-form

speech by segmenting it [15, 16]. These approaches often require additional modules, such as a voice activity detector (VAD). More recent research has aimed to improve the robustness of Transformer-based models for long-form speech without the need for segmentation by enhancing attention mechanisms [17]. However, these studies typically involve modifying or redesigning model architectures, making it challenging to leverage existing pre-trained models.

In long-form speech recognition, while CTC-only models perform well, the performance of hybrid models degrades significantly. To address this, we propose a modification to cross-attention that improves the robustness of AED models for long-form speech. During inference, alignment is achieved by utilizing the position corresponding to the maximum CTC prefix score along the encoder output dimension. This approach demonstrates effective across Transformer, Conformer, and E-Branchformer encoders.

2. Related works

ASR datasets like LibriSpeech [18] primarily consist of short utterances under 30 seconds, but real-world applications often involve longer speech segments. To handle long-form speech, E2E ASR models typically segment audio using methods such as VAD or language models [15, 16, 19], or modify the inference process to move the window during transcription [20]. Another approach focuses on architectural improvements, with several studies suggesting encoder modifications [14, 21].

Ensuring monotonic alignment between speech input and text output is crucial in ASR, leading to proposals for attention mechanism adjustments [22, 23, 24, 25]. While these changes are rarely evaluated on long-form audio, [11] showed that enforcing monotonicity in attention benefits longer speech. Additionally, Fast Conformer [17] enhanced performance by introducing limited context attention and global tokens.

In AED-based or hybrid CTC/Attention-based models, maintaining monotonicity between the encoder output and decoder input is equally important, leading to proposals for modifying cross-attention [26, 27]. These methods can achieve monotonic alignment without being overly dependent on the specific encoder architecture (e.g., Conformer [7], E-Branchformer [8]).

In this paper, we build on the work of [26] by introducing Gaussian masking to the cross-attention mechanism to achieve monotonic alignment. However, to find the alignment position, we do not modify the cross-attention during training; instead, we solely leverage the argmax of the CTC prefix score along the encoder output axis during inference. Our method significantly enhances performance on long-form speech with only minimal code modifications and can also be applied to pre-trained hybrid

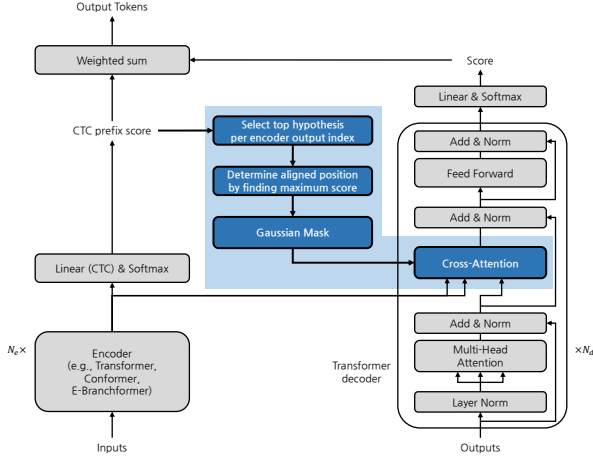


Figure 1: *Inference process of hybrid CTC/attention architecture with proposed method. The blue rounded rectangle denotes our modification, which is applied exclusively during inference. During training, the conventional architecture is used without the proposed method.*

models. Furthermore, as it can be selectively applied only when the speech is long, it maintains strong performance on short utterances as well.

3. Method

In this section, we briefly introduce cross-attention in AED models and then describe methods for positional alignment.

3.1. Cross-attention

The cross-attention mechanism in the Transformer decoder, as shown in Fig. 1, models the relationship between the encoder’s final output and hidden text vectors. Unlike self-attention, in cross-attention, the queries come from the decoder while the keys and values come from the encoder, allowing the decoder to consider the entire encoder sequence at each time-step. Given that $Y = (y_0, \dots, y_i, \dots, y_l)$ denotes the decoder hidden state value for each layer, and $X = (x_0, \dots, x_j, \dots, x_{T-1})$ denotes the final output of the encoder. The corresponding cross-attention weights between the i -th decoder hidden state value and the j -th encoder output can then be expressed as follows:

$$a_{ij} = \text{softmax} \left(\frac{(y_i W^Q)(x_j W^K)^T}{\sqrt{d_k}} \right), \quad 0 \leq i \leq l \quad (1)$$

At each inference time-step l , the above equation is computed for all i ranging from 0 to l . W^Q and $W^K \in \mathbb{R}^{d_{\text{model}} \times d_k}$, where d_{model} is attention dimension and d_k is attention dimension divided by the number of attention head, serve as the linear transformation matrices.

3.2. Cross-attention with positional alignment

In applications like speech recognition, the input and output are strictly aligned in a monotonous order, meaning that only a specific range of input is most relevant to generate the current output token. Therefore, in ASR, it is more effective to focus on a limited range of encoder output in cross-attention, rather than considering the entire input sentence. In a conventional

AED structure, cross-attention can focus on any part of the input. However, by limiting the range of cross-attention, ASR performance can be improved.

3.2.1. Previous work

In [26], a Gaussian mask was added to the conventional cross-attention mechanism to align the positions between the encoder output and the decoder hidden state value as follows:

$$a_{ij} = \text{softmax} \left(\frac{(y_i W^Q)(x_j W^K)^T}{\sqrt{d_k}} + M_{ij} \right) \quad (2)$$

where, M_{ij} represents the Gaussian mask applied to the attention mechanism to adjust the degree of attention, defined as:

$$M_{ij} = \frac{-(j - k)^2}{2\sigma^2} \quad (3)$$

In Eq. 3 the variance of the Gaussian, σ controls the range of index affected by attention mask, and k represents the aligned input position. The value of k , which depends on the decoder hidden state value index i , indicates the most relevant encoder location where cross-attention should focus to compute the output for the i -th token in the decoder. It is crucial to monotonically determine the optimal value of k at each step of generating the output sequence. In [26], the position corresponding to the maximum cross-attention weight was selected as k , σ was treated as a learnable parameter of the network that depends on both the input and target sequences.

3.2.2. Proposed method

In [26], Eq. 3 is required to be applied to both training and inference. By contrast, our work focuses on achieving monotonic attention by searching for alignment positions within the inference process, without relying on these training-based methods [26, 28]. This approach enables a straightforward enhancement of attention, as it does not require fine-tuning a pre-trained model or training a modified model from scratch. Fig. 1 depicts the inference process of the conventional hybrid model and our proposed method.

We revise Eq. 3 into a learnable-parameter-free form by replacing k and σ with $\kappa(i)$ and a constant, respectively, as follows, and apply it exclusively during inference.

$$M_{ij} = \frac{-(j - \kappa(i))^2}{2\sigma^2} \quad (4)$$

The cross-attention weights are computed over all parts of the input without accumulating the influence of previous steps; thus, there is a possibility that the k in Eq. 3 is not monotonic. In contrast, the CTC prefix score, calculated at each time-step, accumulates probabilities from previous steps, causing the alignment position to increase or remain unchanged, thereby ensuring its monotonicity. Since the CTC prefix score provides a score for every element in the encoder output X , a higher score indicates a higher likelihood that the corresponding hypothesis will be adopted at the current time-step. These properties are analogous to the cross-attention mechanism’s focus on the most relevant encoder output position, making it a suitable replacement for the k . Consequently, we define $\kappa(i)$ as a function that returns the encoder output index of the hypothesis that achieves the maximum CTC prefix score at each i . In other words,

$$\kappa(i) = \begin{cases} 0, & i = 0 \\ \arg \max_{0 \leq j \leq T-1} (\log p^{\text{ctc}}(h^l, \dots | X)), & i \geq 1 \end{cases} \quad (5)$$

where T denotes the length of the encoder output and $\log p^{\text{ctc}}(h^l, \dots | X)$ is the CTC prefix score [4, 29]. For each encoder output index j , we select the highest-scoring hypotheses among those available and then obtain the aligned position by taking the arg max with respect to j among those selected hypotheses. In the actual beam search used for inference, multiple beams are maintained at each time-step, so $\kappa(i)$ is returned once per beam for each time-step, and these values are used in Eq 4. Initially, $\kappa(i)$ is set to 0.

In Eq. 4, the variance parameter σ determines how wide the attention span is, centered around $\kappa(i)$. We fixed σ at 20, and additional experiments showed that this value does not significantly impact overall performance.

Furthermore, in [26], Eq. 3 was applied only to the lower decoder layers. In contrast, we applied Eq. 4 to every layer, and this configuration achieved the best performance. This discrepancy may be attributed to the fact that we did not apply Eq. 3 during training.

4. Experiments

4.1. Setup

We use the LibriSpeech corpus, which consists of approximately 1,000 hours of English speech, and train Transformer, Conformer, and E-Branchformer models—each with a Transformer decoder—based on the ESPnet [30] toolkit’s `egs2/librispeech/asr1` recipe. Except for the absence of a language model, the configuration is identical to that in the recipe; therefore, all models are trained as conventional hybrid architectures. For the training set, we use the original utterances without any concatenation. This is intended to verify that even without additional techniques, our proposed method alone enables a hybrid model to achieve robust performance in long-form ASR.

Furthermore, since a publicly available implementation of [26] does not exist, we re-implemented it using the same recipe as our approach. The configuration is largely identical to ours, except that σ is treated as a learnable parameter, and the method for obtaining the aligned position k differs. Unlike [26], the encoder is trained with a Conformer instead of a Transformer.

4.2. Baseline results

As reported in [14], Conformer with hybrid architecture exhibits poor recognition performance on long-form speech. To investigate this, we evaluated recognition performance on the LibriSpeech evaluation set across different utterance lengths, as well as on a set created by concatenating three utterances from

Table 1: WER (%) comparison by speech length for Conformer on LibriSpeech. (Hybrid: Hybrid CTC/Attention architecture, CTC: CTC-only architecture)

Len (sec)	test-clean		test-other	
	Hybrid	CTC	Hybrid	CTC
0-5	3.18	3.85	6.80	8.62
5-10	2.42	2.88	5.61	6.62
10-15	2.00	2.69	4.81	5.87
15-20	2.24	2.80	4.61	5.79
20-25	2.11	2.41	4.71	4.98
25-30	8.08	4.49	11.63	5.07
30-35	15.01	4.90	22.26	3.16

Table 2: WER (%) comparison by speech length for Conformer on the concatenated LibriSpeech (Hybrid: Hybrid CTC/Attention architecture, CTC: CTC-only architecture)

Len (sec)	test-clean		test-other	
	Hybrid	CTC	Hybrid	CTC
20-25	3.43	3.87	9.57	11.39
25-30	4.40	3.80	12.18	11.47
30-35	19.14	3.09	26.10	10.73
35-40	47.82	3.88	50.61	10.40
40-45	55.60	2.88	56.60	10.24
45-50	57.53	3.78	55.85	9.34

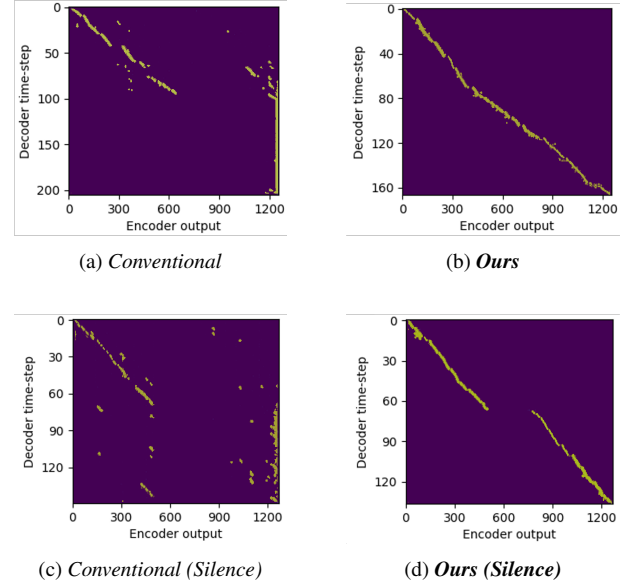


Figure 2: Cross-attention weights of the Conformer from the last encoder layer on approximately 50 seconds of speech. Silence refers to the results from approximately 50 seconds of audio, which includes 10 seconds of silence in the middle. (The image has been revised to better visualize the attention mechanism.)

each set. Utterances longer than 50 seconds were excluded from the experiments. Tables 1 and 2 present the word error rate (WER) of the Conformer model for each utterance length.

When the CTC weight is set to 1.0 (CTC-only), performance degradation does not occur on long-form speech. However, in the hybrid model, WER sharply increases for speech exceeding 25 seconds. We hypothesized that this degradation is due to improper cross-attention alignment. Fig. 2a visualizes the cross-attention weights for a randomly selected long utterance of approximately 50 seconds, revealing a non-diagonal pattern. The WER for this utterance is 51.45%, indicating significant performance degradation.

4.3. Results on proposed method

Table 3 presents the performance of the Conformer with the conventional hybrid architecture alongside that of the Conformer incorporating the proposed method. For short utterances, our approach performs comparably to the baseline, while for speech longer than 20 seconds, it reduces the WER. Our method exhibits a notably stronger performance on long-form

Table 3: WER (%) comparison by speech length for Conformer with proposed method on LibriSpeech. Compared to the hybrid in Table 1, ERR (%) is shown in parentheses.

Len (sec)	test-clean		test-other	
	Ours	CTC	Ours	CTC
0-5	3.18 (0.00)	3.85 (-21.07)	6.92 (-1.76)	8.62 (-26.76)
5-10	2.43 (-0.41)	2.88 (-19.01)	5.65 (-0.71)	6.62 (-18.00)
10-15	2.04 (-2.00)	2.69 (-34.50)	4.98 (-3.53)	5.87 (-22.04)
15-20	2.32 (-3.57)	2.80 (-25.00)	4.89 (-6.07)	5.79 (-25.60)
20-25	2.14 (-1.42)	2.41 (-14.22)	4.12 (12.53)	4.98 (-5.73)
25-30	3.51 (56.56)	4.49 (44.43)	4.13 (64.49)	5.07 (56.41)
30-35	4.46 (70.29)	4.90 (67.36)	1.83 (91.78)	3.16 (85.80)

Table 4: WER (%) comparison by speech length for Conformer with proposed method on the concatenated LibriSpeech. Compared to the hybrid in Table 2, ERR (%) is shown in parentheses.

Len (sec)	test-clean		test-other	
	Ours	CTC	Ours	CTC
20-25	3.29 (4.08)	3.87 (-12.83)	9.53 (0.42)	11.39 (-19.02)
25-30	3.08 (30.00)	3.80 (13.64)	9.83 (19.29)	11.47 (5.83)
30-35	2.41 (87.41)	3.09 (83.86)	9.51 (63.56)	10.73 (58.89)
35-40	3.33 (93.04)	3.88 (91.89)	8.97 (82.28)	10.40 (79.45)
40-45	2.70 (95.14)	2.88 (94.82)	10.06 (82.23)	10.24 (81.91)
45-50	3.69 (93.59)	3.78 (93.43)	8.41 (84.94)	9.34 (83.28)

speech. In the 40-45-second range, it achieves an error reduction rate (ERR) of up to 95.14% (55.60% vs. 2.70%). Detailed results are provided in Table 4.

Fig. 2b illustrates the cross-attention weights after applying our proposed method to the same utterance depicted in Fig. 2a. The attention after applying our proposed method shows a diagonal pattern, and the recognition performance is enhanced with an ERR of 91.55% (51.45% vs. 4.35%), validating our hypothesis that cross-attention misalignment is the cause of performance degradation in long-form speech with hybrid architecture. Furthermore, our method remains effective even in the presence of silence. When a 10-second silence was inserted between two randomly selected 20-second utterances, ERR improved to 95.74% (42.73% vs. 1.82%). Figs. 2d and 2c illustrate the cross-attention for that utterance.

Since our method is applied only during inference, it enables the selective application of our positional alignment to the cross-attention mechanism exclusively for long-form speech inputs. Based on our results, where our method shows an effect for utterances over 20 seconds, we used the conventional structure for those under 20 seconds and our method for those over 20 seconds. We achieved performance improvements on both LibriSpeech and concatenated LibriSpeech. Furthermore, due to the flexibility of our method, it can be easily adapted to other AED-based ASR models. We applied our approach to Transformer and E-Branchformer models and observed similar performance improvements. Detailed results are presented in Tables 5 and 6.

In the case of [26], the performance on short-form speech is comparable to or slightly improved over the baseline. The performance on long-form speech is also improved; however, it remains significantly lower than our proposed method. As shown in Figs. 2a and 2c, cross-attention in long-form speech exhibits alignment errors. From this, it can be inferred that the position of the maximum cross-attention weight, as proposed in [26], is not the optimal aligned position for long-form speech and is the cause of its lower performance. In contrast, our method can readily obtain alignments for long-form speech without re-

Table 5: WER (%) comparison on LibriSpeech. (Hybrid: Hybrid CTC/Attention architecture. CTC (CTC-only) and Ours (Proposed method) are applied only to utterances longer than 20 seconds.)

Model	dev-clean	dev-other	test-clean	test-other
Hybrid				
Conformer	2.45	5.67	2.74	5.85
Transformer	3.12	8.09	3.45	7.99
E-Branchformer	1.99	4.52	2.17	4.72
CTC				
Conformer	2.21	5.60	2.53	5.57
Transformer	3.17	8.15	3.44	8.10
E-Branchformer	1.94	4.54	2.15	4.61
Previous work [26]				
Conformer	2.41	5.71	2.48	5.66
Ours				
Conformer	2.17	5.54	2.48	5.52
Transformer	3.06	7.97	3.28	7.91
E-Branchformer	1.92	4.51	2.10	4.58

Table 6: WER (%) comparison on concatenated LibriSpeech. (Hybrid: Hybrid CTC/Attention architecture. CTC: CTC-only, Ours: Proposed method)

Model	dev-clean	dev-other	test-clean	test-other
Hybrid				
Conformer	22.67	28.32	26.41	27.81
Transformer	15.93	24.77	17.95	24.33
E-Branchformer	13.64	18.19	15.51	18.02
CTC				
Conformer	3.37	11.58	3.57	10.86
Transformer	7.21	19.93	7.40	19.68
E-Branchformer	2.62	8.67	2.73	8.27
Previous work [26]				
Conformer	18.23	23.59	20.81	23.98
Ours				
Conformer	3.14	10.30	3.06	9.49
Transformer	5.54	15.92	5.68	15.20
E-Branchformer	2.52	7.73	2.62	7.31

quiring additional fine-tuning or training a modified model from scratch, thereby achieving monotonic cross-attention and superior performance.

5. Conclusion

In this work, we propose a method for improving long-form speech recognition by integrating a Gaussian mask into the Transformer decoder’s cross-attention mechanism during inference. Specifically, we utilize the position index, where the CTC prefix score is maximized along the encoder output index, as the aligned position for constructing the Gaussian mask. In evaluating Conformer models with a hybrid CTC/attention architecture and those with a CTC-only structure across different utterance lengths, we observed notable performance degradation in the hybrid model for utterances of 25-30 seconds or longer. Our method maintains the performance of hybrid models on short utterances while improving performance on longer utterances, outperforming CTC-only models. Furthermore, our method remains effective even when the hybrid model’s encoder is either a Transformer or an E-Branchformer.

6. Acknowledgements

This work was supported by Institute of Information & communications Technology Planning & Evaluation (IITP) grant funded by the Korea government(MSIT)(RS-2022-II220989(2022-0-00989), Development of Artificial Intelligence Technology for Multi-speaker Dialog Modeling)

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